

Routing Protocols and Concepts

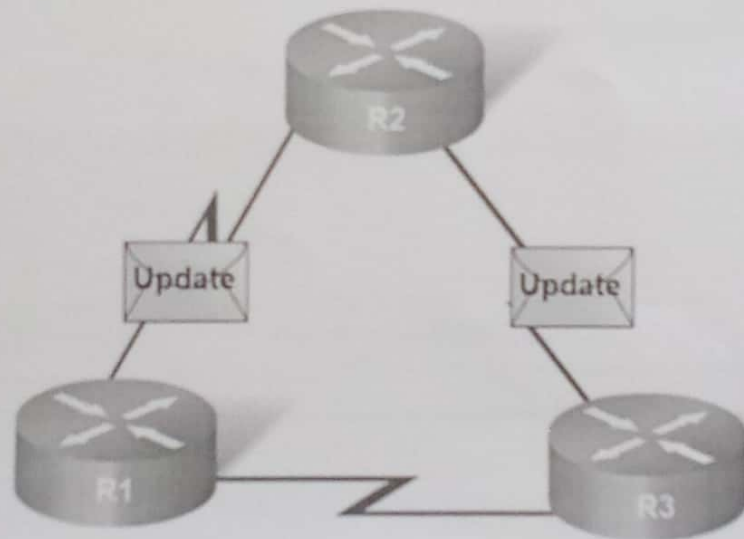
Objectives

- Describe the role of dynamic routing protocols and place these protocols in the context of modern network design.
- Identify several ways to classify routing protocols.
- Describe how metrics are used by routing protocols and identify the metric types used by dynamic routing protocols.
- Determine the administrative distance of a route and describe its importance in the routing process.
- Identify the different elements of the routing table.

Dynamic Routing Protocols

- Function(s) of Dynamic Routing Protocols:
 - Dynamically share information between routers.
 - Automatically update routing table when topology changes.
 - Determine best path to a destination.

Routers Dynamically Pass Updates

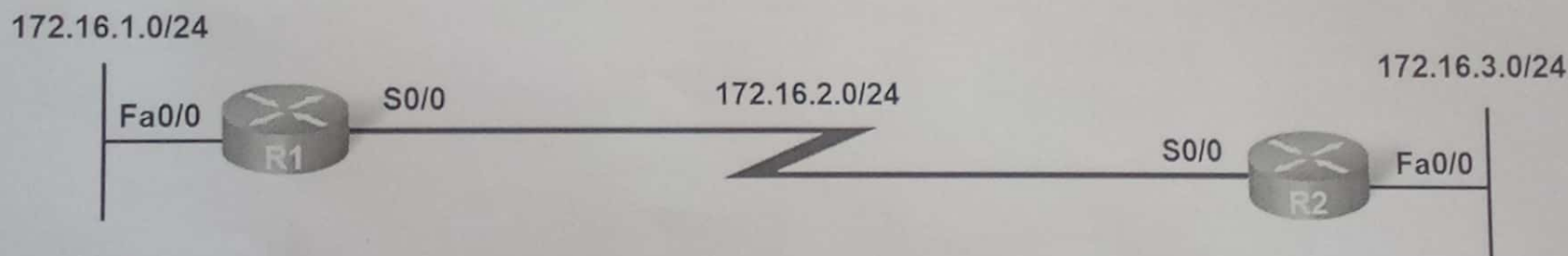


Dynamic Routing Protocols

- The **purpose of a dynamic routing protocol** is to:
 - Discover remote networks
 - Maintaining up-to-date routing information
 - Choosing the best path to destination networks
 - Ability to find a new best path if the current path is no longer available

Routing Protocol Operation

Routing protocols are used to exchange routing information between the routers.



Dynamic Routing Protocols

▪ Components of a routing protocol

– Algorithm

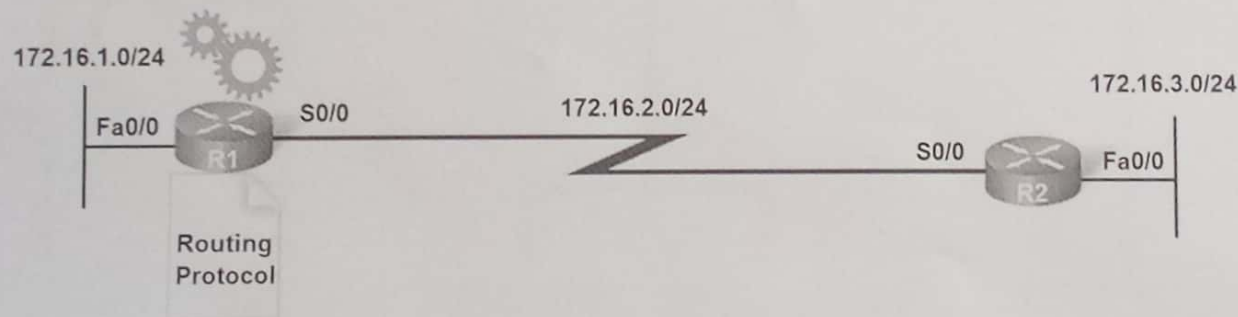
- In the case of a routing protocol algorithms are used for facilitating routing information and best path determination

– Routing protocol messages

- These are messages for discovering neighbors and exchange of routing information

Routing Protocol Operation

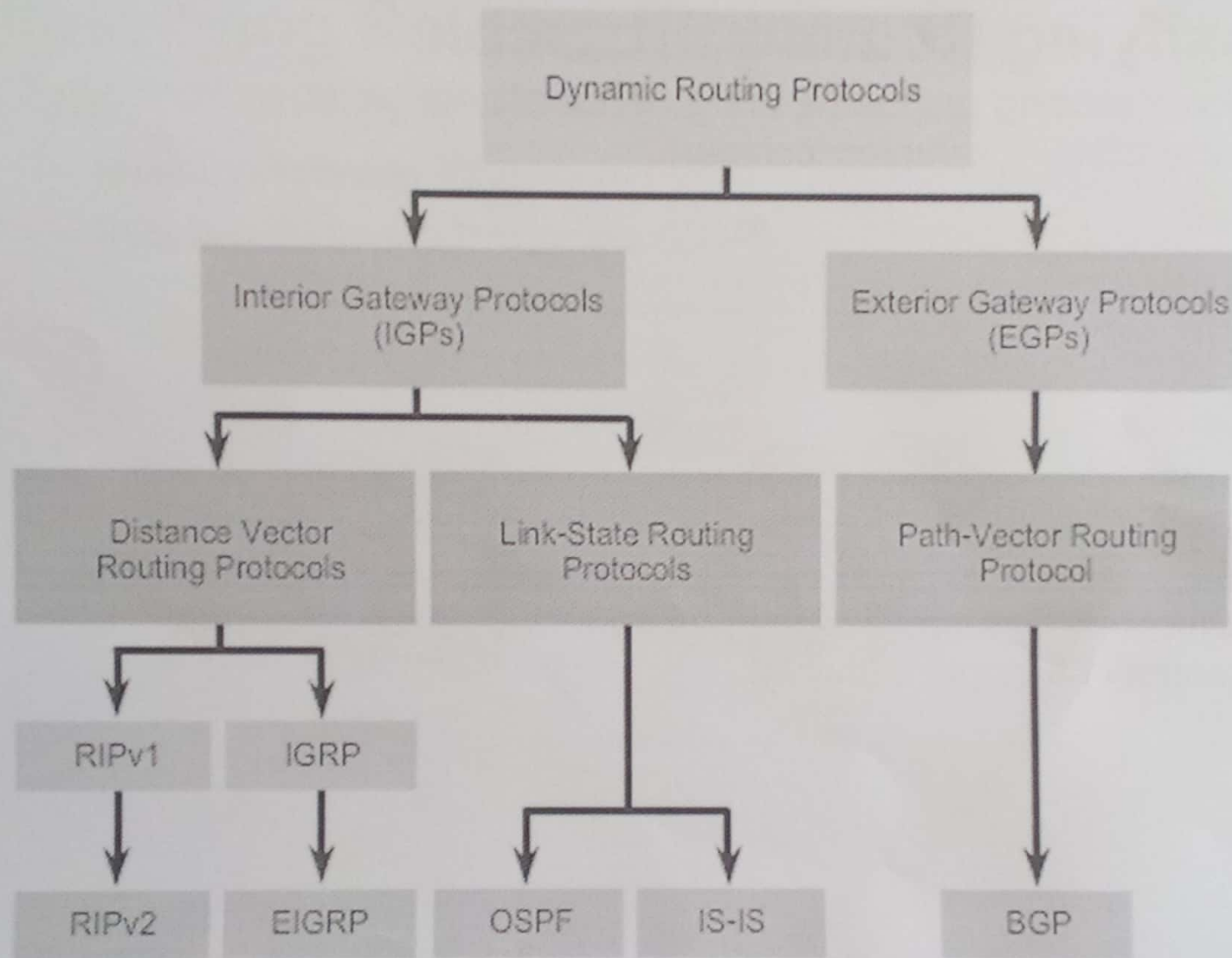
Routing protocols are used to exchange routing information between the routers.



Dynamic Routing Protocols

- Advantages of **static routing**
 - It can backup multiple interfaces/networks on a router
 - Easy to configure
 - No extra resources are needed
 - More secure
- Disadvantages of **static routing**
 - Network changes require manual reconfiguration
 - Does not scale well in large topologies

Routing Protocols Classification



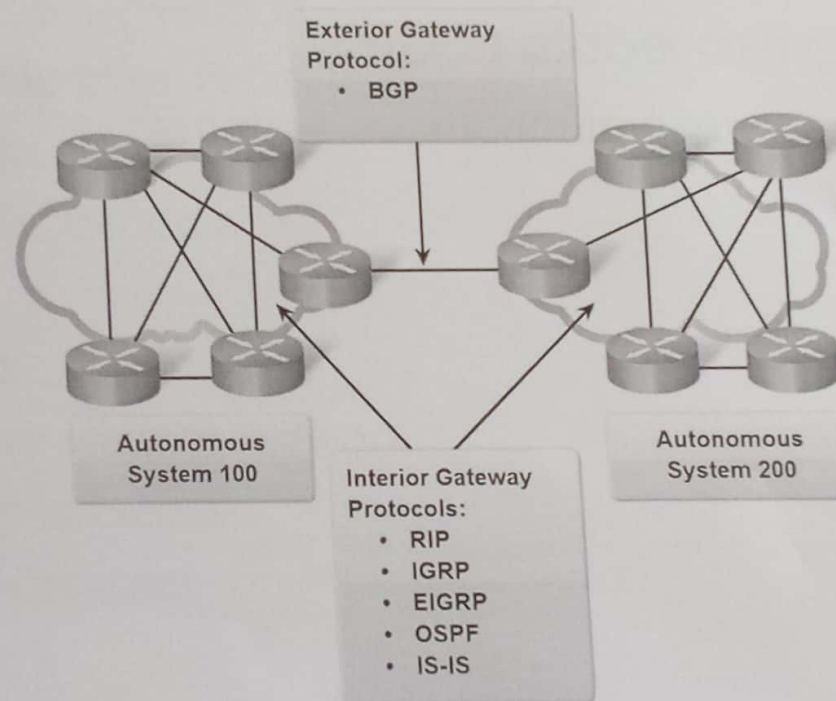
Classifying Routing Protocols

- **Dynamic routing protocols** are grouped according to characteristics. Examples include:
 - RIP
 - IGRP
 - EIGRP
 - OSPF
 - IS-IS
 - BGP
- **Autonomous System** is a group of routers and networks under the control of a single authority.

Classifying Routing Protocols

- Types of routing protocols:
 - Interior Gateway Protocols (IGP)
 - Exterior Gateway Protocols (EGP)

IGP vs. EGP Routing Protocols



Classifying Routing Protocols

▪ Interior Gateway Routing Protocols (IGP)

- Used for routing inside an autonomous system & used to route within the individual networks themselves
- Examples: RIP, EIGRP, OSPF

▪ Exterior Routing Protocols (EGP)

- Used for routing between autonomous systems
- Example: BGPv4

Classifying Routing Protocols

▪ IGP: Comparison of Distance Vector & Link State Routing Protocols

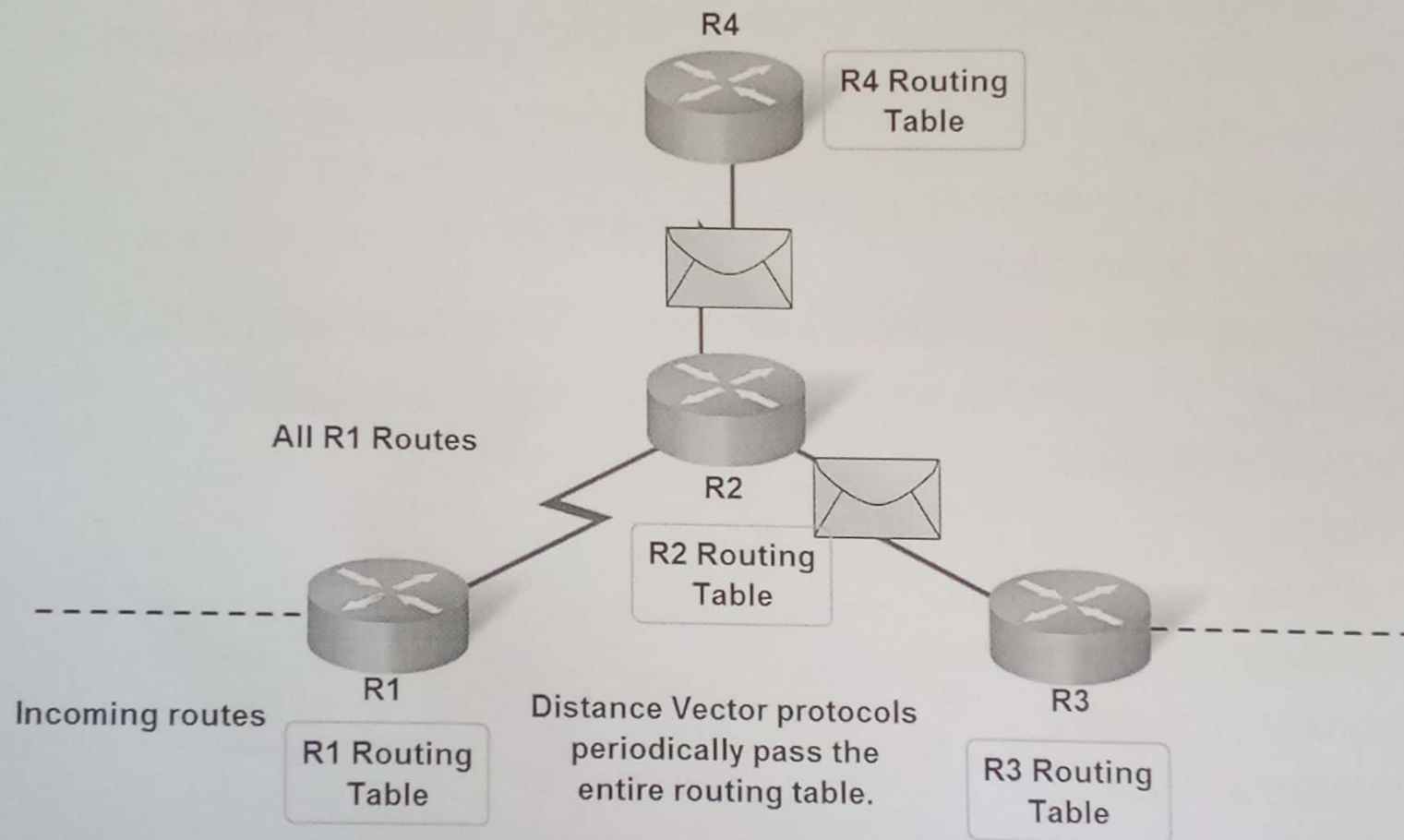
Distance vector

- Routes are advertised as vectors of distance & direction
- Incomplete view of network topology
- Generally, periodic updates

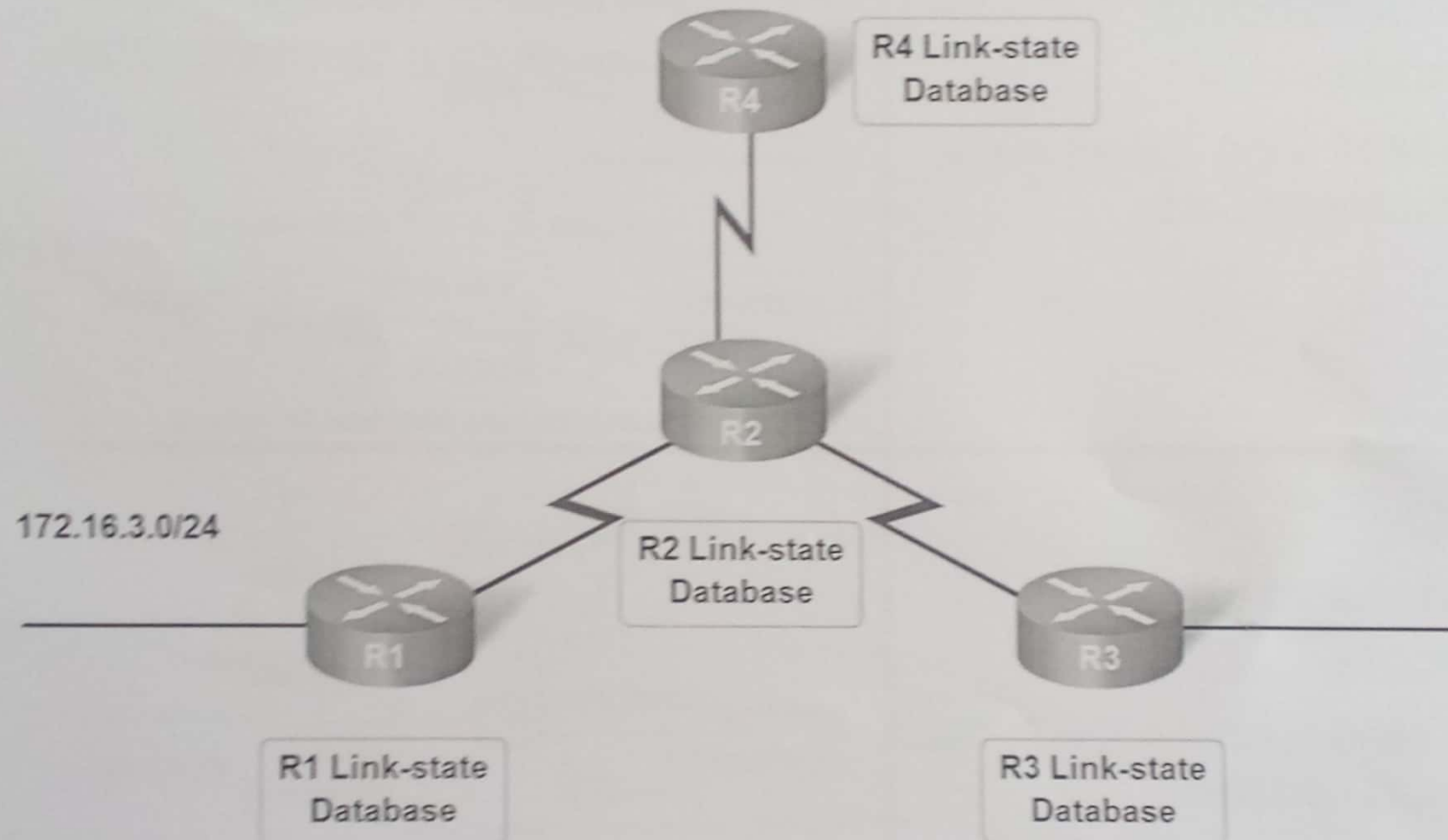
Link state

- Complete view of network topology is created
- Updates are not periodic

Distance Vector Protocol Operation



Link-state Protocol Operation



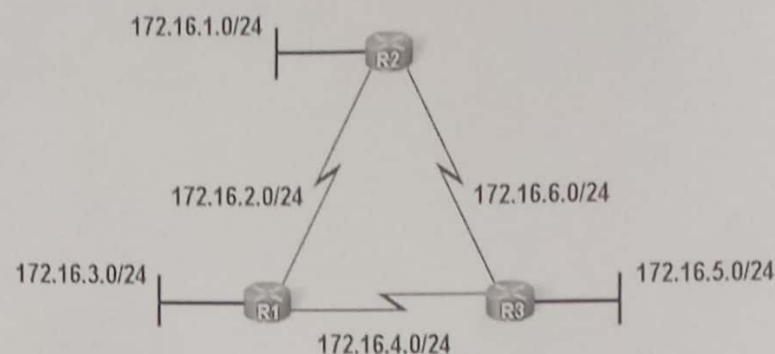
Link-state protocols pass updates when a link's state changes.

Classifying Routing Protocols

▪ Classful routing protocols

- Do NOT send subnet mask in routing updates

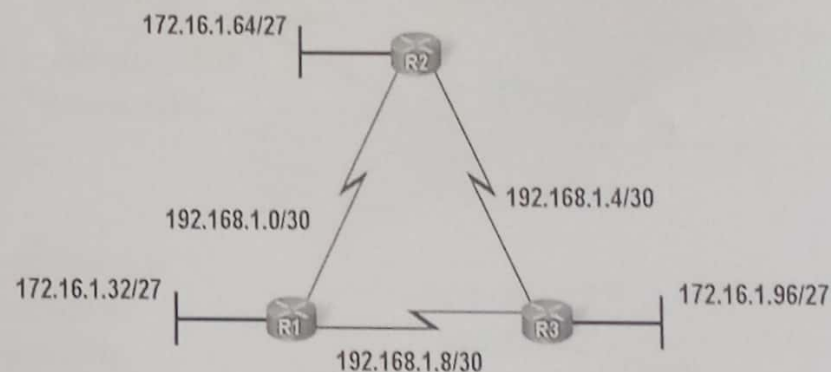
Classful vs. Classless Routing



Classful: Subnet mask is the same throughout the topology

▪ Classless routing protocols

- Do send subnet mask in routing updates

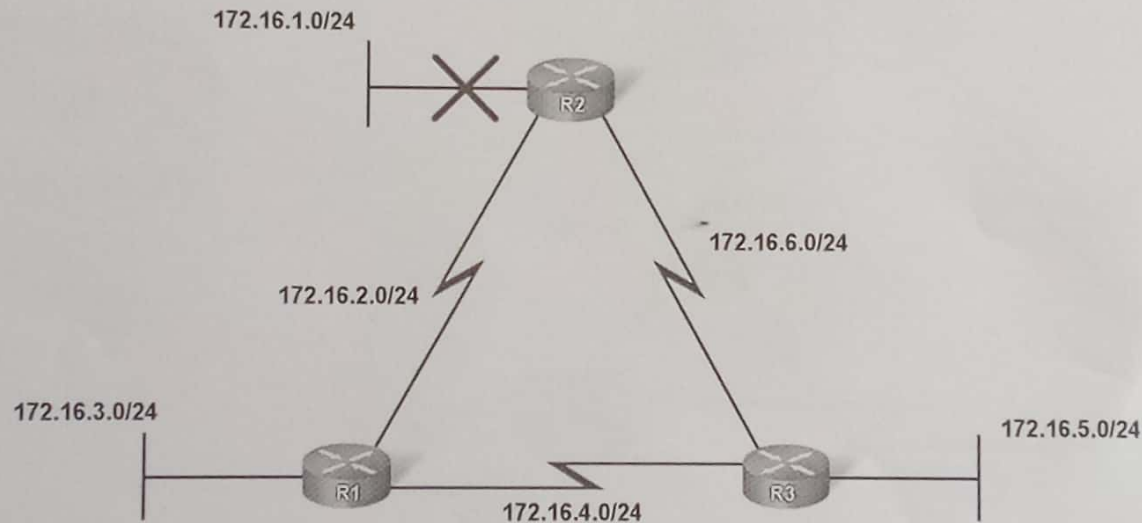


Classless: Subnet mask can vary in the topology

Classifying Routing Protocols

- **Convergence** is defined as when all routers' routing tables are at a state of consistency

Comparing Convergence



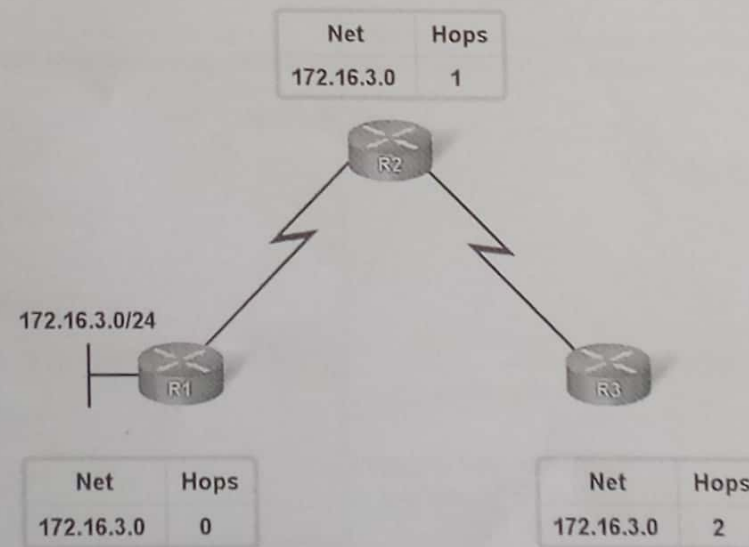
Slower Convergence: RIP and IGRP
Faster Convergence: EIGRP and OSPF

Routing Protocols Metrics

▪ Metric

- A value used by a routing protocol to determine which routes are better than others

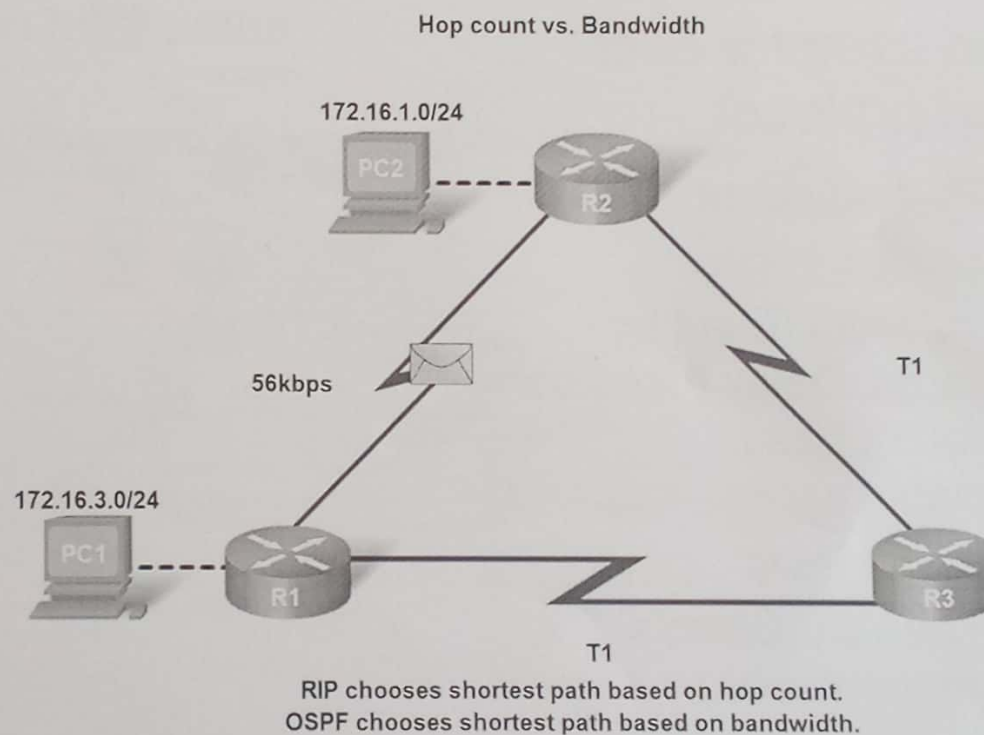
Metrics



Routing Protocols Metrics

▪ Metrics used in IP routing protocols

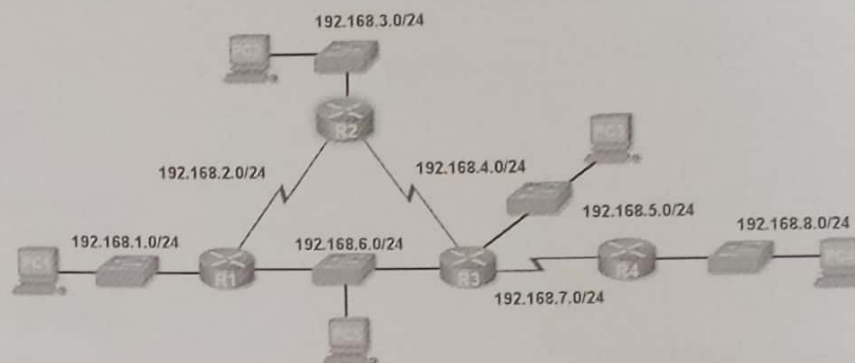
- Bandwidth
- Cost
- Delay
- Hop count
- Load
- Reliability



Routing Protocols Metrics

- The Metric Field in the Routing Table
- **Metric** used for each routing protocol
 - RIP - hop count
 - IGRP & EIGRP - Bandwidth (used by default), Delay (used by default), Load, Reliability
 - IS-IS & OSPF - Cost, Bandwidth (Cisco's implementation)

Metric in the Routing Table



```
R2#show ip route
<output omitted>

Gateway of last resort is not set

R   192.168.1.0/24 [120/1] via 192.168.2.1, 00:00:24, Serial0/0
C   192.168.2.0/24 is directly connected, Serial0/0
C   192.168.3.0/24 is directly connected, FastEthernet0/0
C   192.168.4.0/24 is directly connected, Serial0/1
R   192.168.5.0/24 [120/1] via 192.168.4.1, 00:00:26, Serial0/1
R   192.168.6.0/24 [120/1] via 192.168.2.1, 00:00:24, Serial0/0
    [120/1] via 192.168.4.1, 00:00:26, Serial0/1
R   192.168.7.0/24 [120/1] via 192.168.4.1, 00:00:26, Serial0/1
R   192.168.8.0/24 [120/2] via 192.168.4.1, 00:00:26, Serial0/1
```

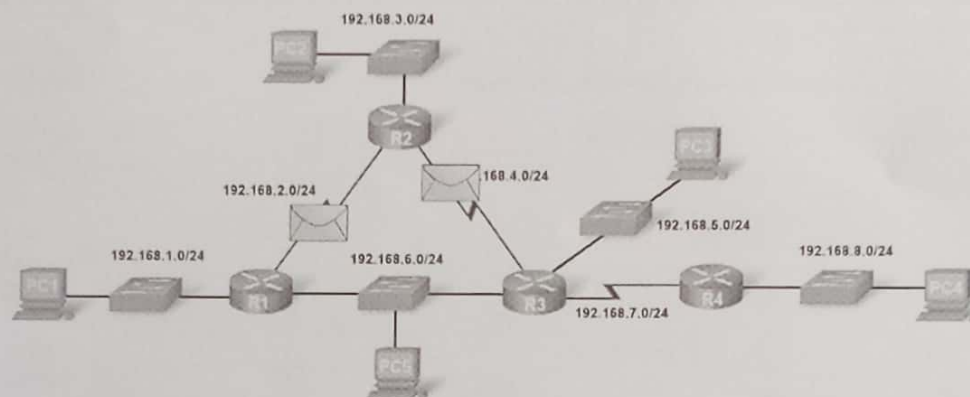
It is 2 hops from R2 to 192.168.8.0/24

Routing Protocols Metrics

▪ Load balancing

- This is the ability of a router to distribute packets among multiple same cost paths

Load Balancing Across Equal Cost Paths



```
R2#show ip route
<output omitted>

R    192.168.6.0/24 [120/1] via 192.168.2.1, 00:00:24, Serial0/0/0
      [120/1] via 192.168.4.1, 00:00:26, Serial0/0/1
```

Administrative Distance of a Route

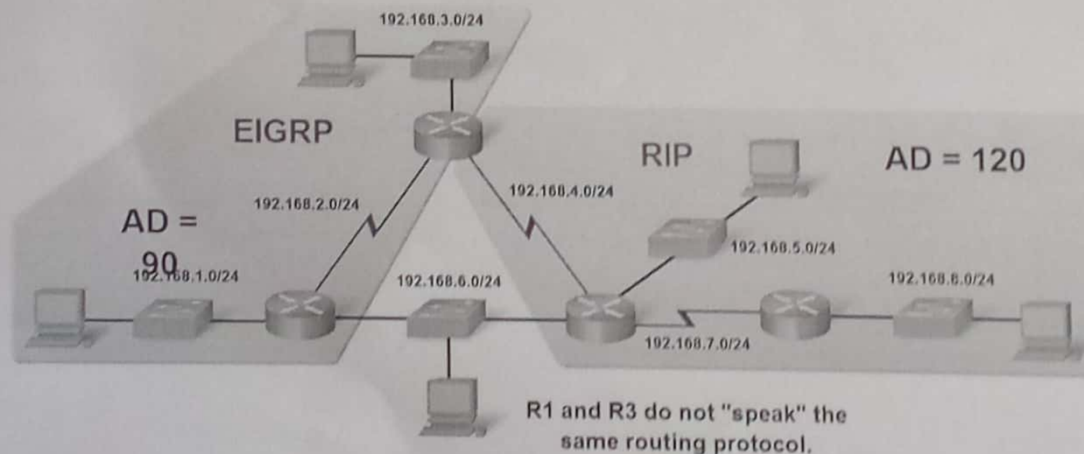
▪ Purpose of a metric

- It's a calculated value used to determine the best path to a destination

▪ Purpose of Administrative Distance

- It's a numeric value that specifies the preference of a particular route

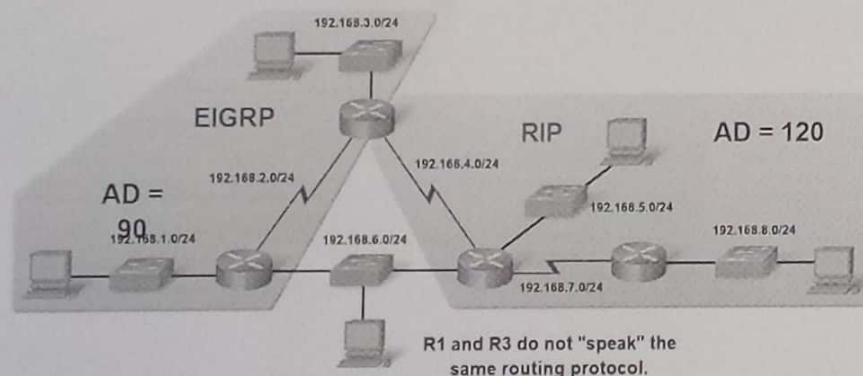
Comparing Administrative Distances



Administrative Distance of a Route

- Identifying the Administrative Distance (AD) in a routing table
 - It is the first number in the brackets in the routing table

Comparing Administrative Distances



```
R2#show ip route
(**output omitted**)
```

Gateway of last resort is not set

```
D 192.168.1.0/24 [90/2172416] via 192.168.2.1, 00:00:24, Serial0/0/0
C 192.168.2.0/24 is directly connected, Serial0/0/0
C 192.168.3.0/24 is directly connected, FastEthernet0/0
C 192.168.4.0/24 is directly connected, Serial0/0/1
R 192.168.5.0/24 [120/1] via 192.168.4.1, 00:00:08, Serial0/0/1
D 192.168.6.0/24 [90/2172416] via 192.168.2.1, 00:00:24, Serial0/0/0
R 192.168.7.0/24 [120/1] via 192.168.4.1, 00:00:08, Serial0/0/1
R 192.168.8.0/24 [120/2] via 192.168.4.1, 00:00:08, Serial0/0/1
```

```
R2#show ip rip database
```

```
192.168.3.0/24    directly connected, FastEthernet0/1
192.168.4.0/24    directly connected, Serial0/0/1
192.168.5.0/24
    [1] via 192.168.4.1, Serial0/0/1
192.168.6.0/24
    [1] via 192.168.4.1, Serial0/0/1
192.168.7.0/24
    [1] via 192.168.4.1, Serial0/0/1
192.168.8.0/24
    [2] via 192.168.4.1, Serial0/0/1
```

Administrative Distance of a Route

▪ Dynamic Routing Protocols

Route Source	Administrative Distance
Connected	0
Static	1
EIGRP summary route	5
External BGP	20
Internal EIGRP	90
IGRP	100
OSPF	110
IS-IS	115
RIP	120
External EIGRP	170
Internal BGP	200

Administrative Distance of a Route

- **Directly connected routes**
 - Have a default **AD** of 0
- **Static Routes**
 - Administrative distance of a static route has a **default value of 1**

```
R2#show ip route 172.16.3.0
Routing entry for 172.16.3.0/24
Known via "static", distance 1, metric 0 (connected)
Routing Descriptor Blocks:
* directly connected, via Serial0/0/0
  Route metric is 0, traffic share count is 1
```

Administrative Distance of a Route

▪ Directly connected routes

- Immediately appear in the routing table as soon as the interface is configured

```
R2#show ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

172.16.0.0/24 is subnetted, 3 subnets

```
C      172.16.1.0 is directly connected, FastEthernet0/0
C      172.16.2.0 is directly connected, Serial0/0/0
S      172.16.3.0 is directly connected, Serial0/0/0
C      192.168.1.0/24 is directly connected, Serial0/0/1
S      192.168.2.0/24 [1/0] via 192.168.1.1
```

Summary

- **Dynamic routing protocols** fulfill the following functions
 - Dynamically share information between routers
 - Automatically update routing table when topology changes
 - Determine best path to a destination
- **Routing protocols are grouped as either**
 - Interior gateway protocols (IGP)**Or**
 - Exterior gateway protocols(EGP)
- **Types of IGPs include**
 - Classless routing protocols - these protocols include subnet mask in routing updates
 - Classful routing protocols - these protocols do not include subnet mask in routing update

Summary

- **Metrics** are used by dynamic routing protocols to calculate the best path to a destination
- **Administrative distance** is an integer value that is used to indicate a router's "trustworthiness"
- **Components of a routing table** include:
 - Route source
 - Administrative distance
 - Metric

Handwritten notes:

CIDR addresses

A / 18
B / 16
C / 24

First octet

	First octet	mask
Class A	1 - 126	N.N.N.H
B	128 - 191	N.N.N.H
C	192 - 223	N.N.N.H
D	224 - 255	

(1 sub block 127.0.0.1) designated